

Source Documentation

Global Game Jam 2026: Hardware & Software

Contents

1. Introduction	4
1.1. Important Dates and Timeline	4
2. Technology Used	4
2.1. Hardware	4
2.1.1. Raspberry Pi Pico 2 WH	4
2.1.2. Standard Push Buttons	4
2.1.3. 22 Gauge Wire	4
2.1.4. 2.2" LCD Screen	5
2.1.5. Solder Boards	5
2.1.6. Speakers	5
2.1.7. Audio Interface and Amplifier	5
2.1.8. Total Hardware cost	5
2.2. Software	6
2.2.1. Pico GB Project	6
2.2.2. Gb Studio	6
2.2.3. Solidworks/FreeCAD	6
2.2.4. Git	6
2.2.5. Total Software Cost	6
3. Schematics and Models	7
3.1. Printed Shell	7
3.2. Hardware Internals	7
3.3. Git Workflow	7
4. Prior to Day One	7
4.1. Buying the Hardware	7
4.2. Checking Competition Legality	7
4.3. Preparations	7
5. The Game Jam	8
5.1. Theme	8

5.2. Our Game (Initial Premise)	8
5.3. Total Timeline	8
6. Results and Reflection	8

1. Introduction

This documentation serves as the collection of information between the hardware and software of this project.

1.1. Important Dates and Timeline

The entire game was made on a single weekend, from January 30th to February 1st. The hardware construction began 2 weeks earlier. To stay within the nature of the competition, the game can be played, and is presented in a more typical “.exe” format and in a web browser on The Game Jam Website.

2. Technology Used

2.1. Hardware

2.1.1. Raspberry Pi Pico 2 WH

We went with the Raspberry Pi Pico 2W as the “brain” of our demonstration. Inspired by the Pico GB project, we wanted to take the core reverse engineering provided by the project and put our own game and modifications onto the project.

2.1.2. Standard Push Buttons

As a cost saving measure, we got a large pack of standard push buttons as all inputs for the hardware.

2.1.3. 22 Gauge Wire

We used this wire because it was available and thicker than the original gauge wire used by Pico GB, any gauge thicker than 24 will do.

2.1.4. 2.2” LCD Screen

While not the original model used by Pico GB, we matched their specs exactly, 2.2inch diagonally with 220p x 176p resolution.

2.1.5. Solder Boards

To save on weight and thickness we used some donated solderboard with a matching connector layout as a breadboard. This allowed u to match our prototyping, cut cost, and save time when building the final model.

2.1.6. Speakers

We went with an 8ohm resistive speaker for sound

2.1.7. Audio Interface and Amplifier

We went with a standard I2C sound board for all audio interfacing

2.1.8. Total Hardware cost

Item Cost
Raspberry Pi Pico 2 WH \$5.00
Standard Push Buttons (Large bundle) \$0.50
22 Gauge Wire \$some
2.2” LCD Screen (2 pack) \$5.00
Solder Boards (3 pack) \$5.00
Speaker \$5.00
Audio Interface (3 pack) \$5.00
Total Cost \$5.00

Table 1: Total Cost Breakdown

2.2. Software

2.2.1. Pico GB Project

To save time and resources, we used the original firmware from the Pico GB project. We compiled our game jam game into the emulator format to then be displayed and demo'd on.

Link: <https://github.com/YouMakeTech/Pico-GB>

2.2.2. Gb Studio

This was our go-to game engine for designing, testing, and compiling/presenting our game to the game jam community.

Link: <https://www.gbstudio.dev>

2.2.3. Solidworks/FreeCAD

For all 3D models for the Gameboy cases, we used a variety of solidworks/FREECAD to design, view, and slice in order for us to 3D print what we needed. All models are included in section 3.

2.2.4. Git

As seen with this repository, git was the version control system the team agreed on. We then uploaded the repository to github.

Link: <https://git-scm.com>

Link: <https://github.com/JEvan234/Global-Game-Jam-2026>

2.2.5. Total Software Cost

All of the software used for this project was free and open source software (FOSS). Outside of solidworks, which has

alternatives like FREECAD, none of the software has a purchase or liscensing fee.

3. Schematics and Models

3.1. Printed Shell

3.2. Hardware Internals

3.3. Git Workflow

4. Prior to Day One

4.1. Buying the Hardware

We bought our hardware from a variety of storefronts ranging from our local Micro Center to Amazon online.

4.2. Checking Competition Legality

According to the Game Jam Website, link at bottom, All alternative-hardware titles were competiton legal and welcome.

Link: <https://globalgamejam.org/jam-sites/2026/kennesaw-state-university>

4.3. Preperations

Leading into day one of the Game Jam:

- Initialized the repository and documentation
- Acquired materials
- Downloaded software
- Assembled team and established goals for the jam

5. The Game Jam

5.1. Theme

The Game jam Theme this year is

5.2. Our Game (Initial Premise)

5.3. Total Timeline

6. Results and Reflection